NUCLEAR FUEL CYCLE

Nuclear Fuel and its Fabrication

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- Fuel fabrication is the last step in the process of turning uranium into nuclear fuel rods.
- Batched into assemblies, the fuel rods form the majority of a reactor core's structure.
- This transformation from a fungible material – uranium – to high-tech reactor components is conceptually different from the refining and preparation of fossil fuels.
- Nuclear fuel assemblies are specifically designed for particular types of reactors and are made to exacting standards.
- Utilities and fabricators have collaborated to greatly improve fuel assembly performance, and an international accident-tolerant fuel program is under way.
- While all present fuel is oxide, R&D is focused on metal, nitride and other forms. The first modern metal fuel is due to be trialled in commercial reactors.

Nuclear reactors are powered by fuel containing fissile material. The fission process releases large amounts of useful energy and for this reason the fissioning components – U-235 and/or Pu- 239 – must be held in a robust physical form capable of enduring high operating temperatures and an intense neutron radiation

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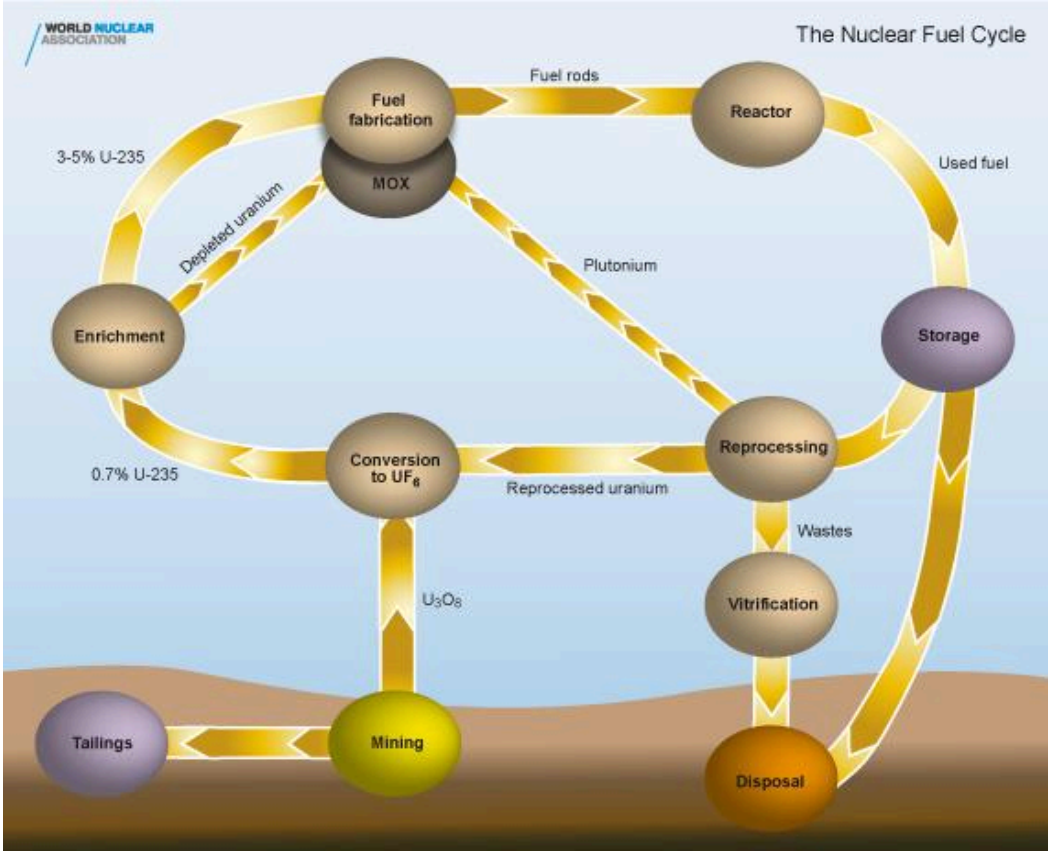
environment. Fuel structures need to maintain their shape and integrity over a period of several years within the reactor core, thereby preventing the leakage of fission products into the reactor coolant.

The standard fuel form comprises a column of ceramic pellets of uranium oxide, clad and sealed into zirconium alloy tubes. For light water reactor (LWR) fuel, the uranium is enriched to various levels up to about 4.8% U-235. Pressurised heavy water reactor (PHWR) fuel is usually unenriched natural uranium (0.7% U-235), although slightly-enriched uranium is also used.

Fuel assembly performance has improved since the 1970s to allow increased burn-up of fuel from 40 GWday/tU to more than 60 GWd/tU. This is correlated with increased enrichment levels from about 3.25% to 5% and the use of advanced burnable absorber designs for PWR, using gadolinium. Core monitoring giving detailed real-time information has enabled better fuel performance also.

The fabrication of fuel structures – called assemblies or bundles – is the last stage of the front end of the nuclear cycle shown in Figure 1, and represents less than 20% of the final cost of the fuel. The process for uranium-plutonium mixed oxide (MOX) fuel fabrication is essentially the same – notwithstanding some specific features associated with handling the plutonium component.

Figure 1: The closed nuclear fuel cycle, showing primary and recycled materials flow



The industry is dominated by four companies serving international demand for light water reactors: Areva, Global Nuclear Fuel (GNF), TVEL and Westinghouse. GNF is mostly for BWR, and TVEL for PWR.

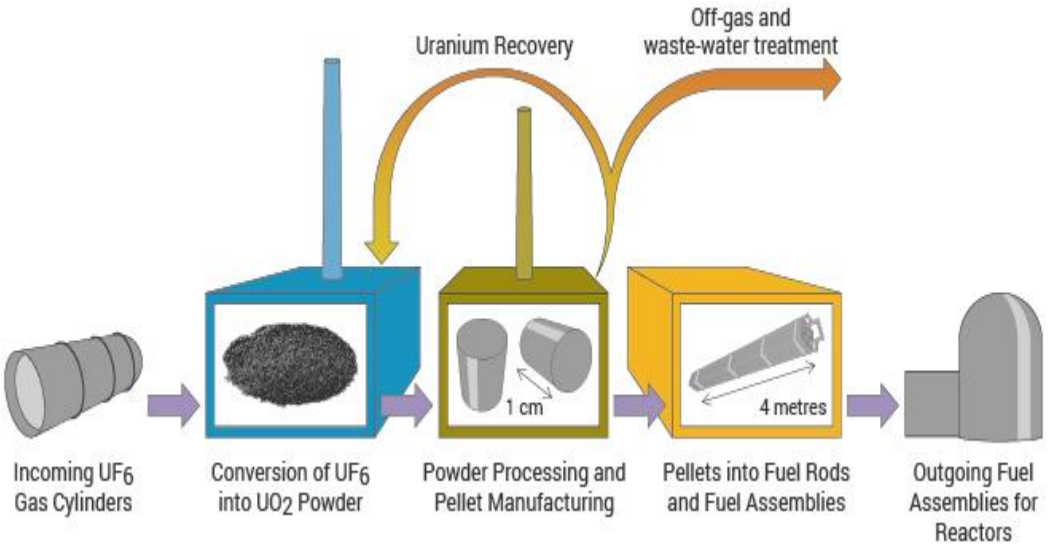
Nuclear fuel fabrication – process overview

There are three main stages in the fabrication of the nuclear fuel structures used in LWRs and PHWRs:

1. Producing pure uranium dioxide (UO_2) from incoming UF_6 or UO_3.
2. Producing high-density, accurately shaped ceramic UO_2 pellets.
3. Producing the rigid metal framework for the fuel assembly – mainly from zirconium alloy; and loading the fuel pellets into the fuel rods, sealing them and assembling the rods into the final fuel assembly structure.

These steps are illustrated in Figure 2.

Figure 2: The fuel fabrication process



UO_2 powder production

Uranium arrives at a fuel manufacturing plant in one of two forms, uranium hexafluoride (UF_6) or uranium trioxide (UO_3), depending on whether it has been enriched or not. It needs to be converted to uranium dioxide (UO_2) prior to pellet fabrication. Most fabrication plants have their own facilities for effecting this chemical conversion (some do not, and acquire UO_2 from plants with excess conversion capacity). Chemical conversion to and from UF_6 are distinct processes, but both involve the handling of aggressive fluorine compounds and plants may be set up to do both.

Conversion to UO_2 can be done using 'dry' or 'wet' processes. In the dry method, UF_6 is heated to a vapour and introduced into a two stage reaction vessel (eg, rotary kiln) where it is first mixed with steam to produce solid uranyl fluoride (UO_2F_2) – this powder moves through the vessel to be reacted with H_2 (diluted in steam) which removes the fluoride and chemically reduces the uranium to a pure microcrystalline UO_2 product.

Wet methods involve the injection of UF_6 into water to form a UO_2F_2 particulate slurry. Either ammonia (NH_3) or ammonium carbonate ($(NH_3)2CO_3$) is added to this mixture and the UO_2F_2 reacts to produce; ammonium diuranate (ADU, $(NH_3)2U_2O_7$) in the first case, or ammonium uranyl carbonate (AUC, $UO_2CO_3.(NH_3)2CO_3$) in the latter case. In both cases the slurry is filtered, dried and heated in a reducing atmosphere to pure UO_2. The morphology of UO_2 powders deriving from the ADU and AUC routes are different, and this has a bearing on final pellet microstructure.

Wet methods are slightly more complex and give rise to more wastes, however, the greater flexibility in terms of UO_2 powder

properties is an advantage.

For the conversion of UO_3 to UO_2, water is added to UO_3 so that it forms a hydrate. This solid is fed (wet or dry) into a kiln operating with a reducing atmosphere and UO_2 is produced.

Manufacture of ceramic UO_2 pellets

The UO_2 powder may need further processing or conditioning before it can be formed into pellets:

- Homogenization: powders may need to be blended to ensure uniformity in terms of particle size distribution and specific surface area.
- Additives: U_3O_8 may be added to ensure satisfactory microstructure and density for the pellets. Other fuel ingredients, such as lubricants, burnable absorbers (e.g. gadolinium) and pore-formers may also need to be added.

Conditioned UO_2 powder is fed into dies and pressed biaxially into cylindrical pellet form using a load of several hundred MPa – this is done in pressing machines operating at high speed. These 'green' pellets are then sintered by heating in a furnace at about 1750°C under a precisely controlled reducing atmosphere (usually argon-hydrogen) in order to consolidate them. This also has the effect of decreasing their volume. The pellets are then machined to exact dimensions – the scrap from which being fed back into an earlier stage of the process. Rigorous quality control is applied to ensure pellet integrity and precise dimensions.

For most reactors pellets are just under one centimetre in diameter and a little more than one centimetre long. A single pellet in a typical reactor yields about the same amount of energy as one tonne of steaming coal.

Burnable absorbers (or burnable 'poisons') such as gadolinium may be incorporated (as oxide) into the fuel pellets of some rods to limit reactivity early in the life of the fuel. Burnable absorbers have a very high neutron absorption cross-section and compete strongly for neutrons, after which they progressively 'burn-out' and convert into nuclides with low neutron absorption leaving fissile (U-235) to react with neutrons. Burnable absorbers enable longer fuel life by allowing higher fissile enrichment in fresh fuel, without excessive initial reactivity and heat being generated in the assembly.

Gadolinium, mostly at up to 3g oxide per kilogram of fuel, requires slightly higher fuel enrichment to compensate for it, and also after burn-up of about 17 GWd/t it retains about 4% of its absorbtive effect and does not decrease further. Zirconium diboride integral fuel burnable absorber (IFBA) as a thin coating on normal pellets burns away more steadily and completely, and has no impact on fuel pellet properties. It is now used in most US reactors and a few in Asia. China has this technology for AP1000 reactors.

MOX pellets – see later section.

Manufacture and loading of the fuel assembly framework

Nuclear fuel designs dictate that the pellet-filled rods have a precise physical arrangement in terms of their lattice pitch (spacing), and their relation to other features such as water (moderator) channels and control-rod channels. The physical structures for holding the fuel rods are therefore engineered with extremely tight tolerances. They must be resistant to chemical corrosion, high temperatures, large static loads, constant vibration, fluid and mechanical impacts. Yet they must also be as neutron-transparent as possible.

Assembly structures comprise a strong framework made from steel and zirconium upon which are fixed numerous grid support pieces that firmly hold rods in their precise lattice positions. These are made from zirconium alloy and must permit (and even enhance) the flow of coolant water around the fuel rod. The grid structures grip the fuel rod and so are carefully designed to minimise the risk of vibration-induced abrasion on the cladding tube – called 'fretting' wear.

All fuel fabricators have highly sophisticated engineering processes and quality control for the timely manufacture of their assembly structures.

Pellets meeting QA specifications are loaded into tubes made from an appropriate zirconium alloy, referred to as the 'cladding'. The filled tube is flushed with helium and pressurized with tens of atmospheres (several MPa) of this gas before the ends are sealed at each end by precision welding. A free space is left between the top of the pellet stack and the welded end-plugs – this is called the 'plenum' space and it accommodates thermal expansion of the pellets and some fission product gases. A spring is usually put into the plenum to apply a compressive force on the pellet stack and prevent its movement.

The completed fuel rods are then fixed into the prefabricated framework structures that hold the rods in a precisely defined grid arrangement.

In order to maximize the efficiency of the fission reaction the cladding and indeed all other structural parts of the assembly must be as transparent as possible to neutrons. Different forms of zirconium alloy, or zircaloy, are therefore the main materials used for cladding. This zircaloy includes small amounts of tin, niobium, iron, chromium and nickel to provide necessary strength and corrosion resistance. Hafnium, which typically occurs naturally with zirconium deposits, needs to be removed because of its high neutron absorption cross-section. The exact composition of the alloy used depends on the manufacturer and is an important determiner in the quality of the fuel assembly. Zircaloy oxidizes in air and water, and therefore it has an oxidized layer which does not impair function.

Safety considerations

Rigorous quality control measures are employed at all process points in order to ensure traceability of all components in case of failures.

The major process safety concerns at nuclear fuel fabrication facilities are those of fluoride handling and the risk of a criticality event if insufficient care is taken with the arrangement of fissile materials. Both risks are managed through the rigorous control of materials, indeed, fuel fabrication facilities operate with a strict limitation on the enrichment level of uranium that is handled in the plant – this cannot be higher than 5% U-235, essentially eliminating the possibility of inadvertent criticality.

Types of nuclear fuel assemblies for different reactors

There is considerable variation among fuel assemblies designed for the different types of reactor. This means that utilities have limited choice in suppliers of fabricated fuel assemblies, especially for PWRs.

PWR fuel

Pressurised water reactors (PWRs) are the most common type of nuclear reactor accounting for two-thirds of current installed nuclear generating capacity worldwide. A PWR core uses normal water as both moderator and primary coolant – this is kept under considerable pressure (about 10 MPa) to prevent it from boiling, and its temperature rises to about 330°C after its upward passage past the fuel. It then goes through massive pipes to a steam generator.

Fuel for western PWRs is built with a square lattice arrangement and assemblies are characterized by the number of rods they contain, typically, 17×17 in current designs. A PWR fuel assembly stands between four and five metres high, is about 20 cm across and weighs about half a tonne. The assembly has vacant rod positions – space left for the vertical insertion of a control rod. Not every assembly position requires fuel or a control rod, and a space may be designated as a "guide thimble" into which a neutron source rod, specific instrumentation, or a test fuel segment can be placed.

A PWR fuel assembly comprises a bottom nozzle into which rods are fixed through the lattice and to finish the whole assembly it is crowned by a top nozzle. The bottom and top nozzles are heavily constructed as they provide much of the mechanical support for the fuel assembly structure. In the finished assembly most rod components will be fuel rods, but some will be guide thimbles, and one or more are likely to be dedicated to instrumentation. A PWR fuel assembly is shown in Figure 3. PWR fuel assemblies are rather uniform compared with BWR ones, and those in any particular reactor must have substantially the same design.

An 1100 MWe PWR core may contain 193 fuel assemblies composed of over 50,000 fuel rods and some 18 million fuel pellets. Once loaded, fuel stays in the core for several years depending on the design of the operating cycle. During refueling, every 12 to 18 months, some of the fuel - usually one third or one quarter of the core – is removed to storage, while the remainder is rearranged to a location in the core better suited to its remaining level of enrichment.

Russian PWR reactors are usually known by the Russian acronym VVER. Fuel assemblies for these are characterized by their hexagonal arrangement, but are otherwise of similar length and structure to other PWR fuel assemblies. Most is made by TVEL in Russia, but Westinghouse in Sweden also fabricates it and is increasing capacity to do so. TVEL is instigating using erbium as a

burnable poison in fuel enriched to about 6.5% in order to prolong the intervals between refuelling to two years.

Figure 3: Schematic view of PWR fuel assembly (Mitsubishi Nuclear Fuel)

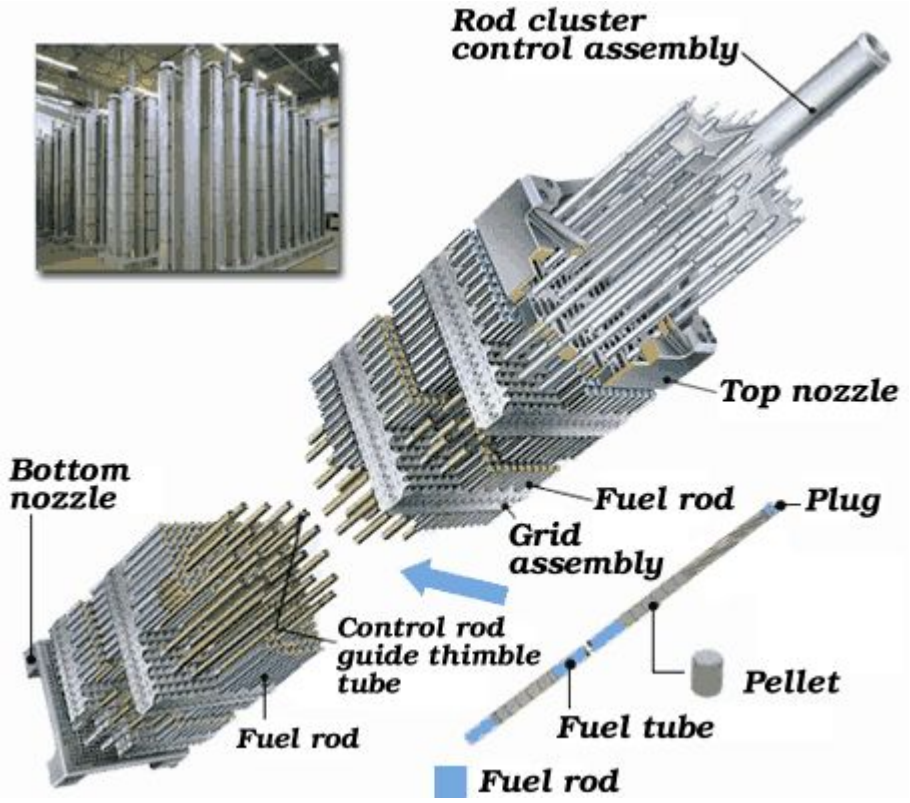
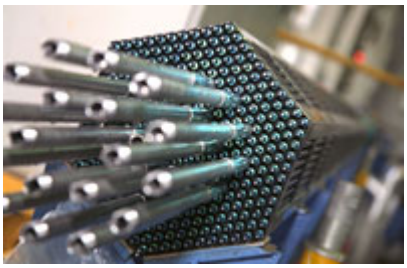


Figure 4: A PWR fuel assembly

Figure 5: A VVER-1000 fuel assembly



BWR fuel

Boiling water reactors (BWRs) are the second most common nuclear reactor type accounting for almost one-quarter of installed nuclear generating capacity. In a boiling water reactor water is turned directly to steam in the reactor pressure vessel at the top of the core and this steam (at about 290°C and 7 MPa) is then used to drive a turbine.

BWRs also use fuel rods comprising zirconium-clad uranium oxide ceramic pellets. Their arrangement into assemblies is again based on a square lattice, with pin geometries ranging from 6x6 to 10x10 or 11x11. Fuel life and management strategy is similar to that for a PWR.

But BWR fuel is fundamentally different from PWR fuel in certain ways: (i) Four fuel assemblies and a cruciform shaped control blade form a 'fuel module', (ii) each assembly is isolated from its neighbours by a water-filled zone in which the cruciform control rod blades travel (they are inserted from the bottom of the reactor), (iii) each BWR fuel assembly is enclosed in a zircaloy sheath or channel box which directs the flow of coolant water through the assembly and during this passage it reaches boiling point, (iv) BWR assemblies contain larger diameter water channels – flexibly designed to provide appropriate neutron moderation in the assembly.

The zircaloy tubes are allowed to fill with water thus increasing the amount of moderator in the central region of the assembly. Different enrichment levels are used in the rods in varying positions – lower enrichments in the outer rods, and higher enrichments near the centre of the bundle. A BWR reactor is designed to operate with 12-15% of the water in the top part of the core as steam, and hence with less moderating effect and thus efficiency there.

For many BWR models, control of reactivity to enable load-following can be achieved by changing the rate of circulation inside the core. Jet pumps located in the annulus between the outer wall of the vessel and an inner wall called the shroud increase the flow

of water up through the fuel assembly. At high flow rates steam bubbles are removed more quickly, and hence moderation and reactivity is increased. When flow rate is decreased, moderation decreases as steam bubbles are present for longer and hence reactivity drops. This allows for a variation of about 25% from the maximum rated power output, enabling load-following more readily than with a PWR.

Control rods are used when power levels are reduced below 75%, but they are not part of the fuel assembly as in a PWR. They are bottom-entry – pushed upwards so that rods intercept the lower, more reactive, zone of the fuel assemblies first.

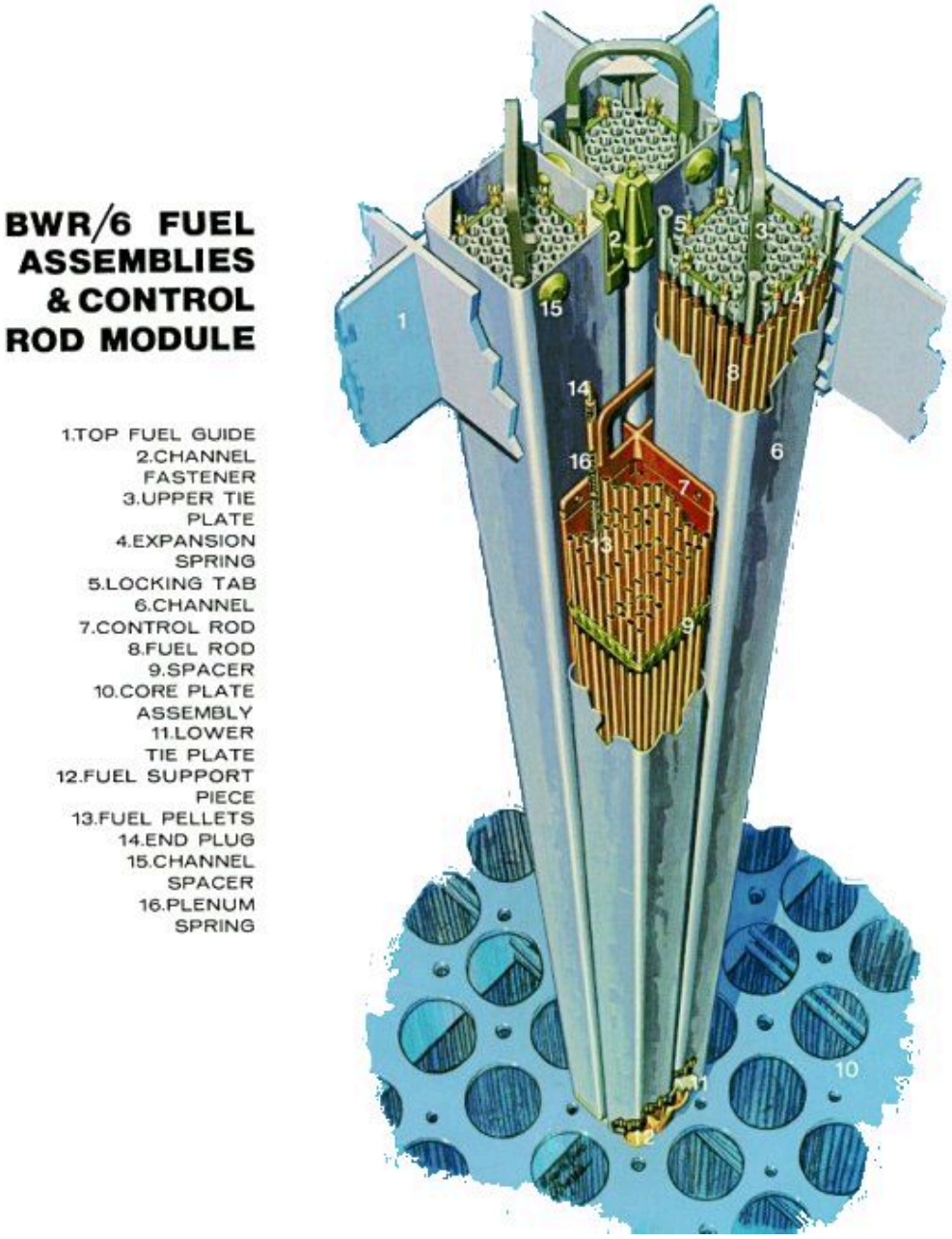
BWR fuel fabrication takes place in much the same way as PWR fuel.

A cross sectional diagram of a BWR assembly is shown in Figure 6. BWR fuel assemblies therefore operate more as individual units, and different designs may be mixed in any core load, giving more flexibility to the utility in fuel purchases.

GE's Global Nuclear Fuels is developing fuel with new clad material – NSF – containing 1% niobium, 1% tin, 0.35% iron (Nb,Sn,Fe) to reduce or eliminate fuel channel distortion due to chemical interaction with zircaloy and in 2013, 8% of cores were using this. Toshiba and ceramics company Ibiden in Japan are developing silicon carbide sheaths or channel boxes for BWR fuel assemblies.

Westinghouse plans to produce lead test assemblies of its TRITON11 fuel (11x11 configuration) for BWRs in 2019. This has a low-tin zirconium channel material and new fuel cladding. It says that this fuel features improved economy, robust mechanical design and high-performing material. It has been optimised for both short- and long-cycle operation as well as for uprated cores and higher burn-ups.

Figure 6: Schematic view of BWR fuel assembly (Nucleartourist and GE)



PHWR (CANDU) fuel

Pressurised heavy water reactors (PHWRs) are originally a Canadian design (also called “CANDU”) accounting for ~6% of world installed nuclear generating capacity. PHWRs use pressure tubes in which heavy water moderates and cools the fuel. They run on natural (unenriched) or slightly-enriched uranium oxide fuel in ceramic pellet form, clad with zirconium alloy.

PHWR fuel rods are about 50 cm long and are assembled into ‘bundles’ approximately 10 cm in diameter. A fuel bundle comprises 28, 37 or 43 fuel elements arranged in several rings around a central axis (see Figure). Their short length means that they do not require the support structures characteristic of other reactor fuel

types. PHWR fuel does not attain high burn-up, nor does it reside in the reactor core for very long and so the fuel pellets swell very little during their life. This means that PHWR fuel rods do not need to maintain a pellet-cladding gap, nor be highly pressurized with a filling gas (as for LWR fuel), indeed, the metal cladding is allowed to collapse onto the fuel pellet thereby assuring good thermal contact.

The fuel bundles are loaded into horizontal channels or pressure tubes which penetrate the length of the reactor vessel (known as the calandria), and this can be done while the reactor is operating at full power. About twelve bundles are loaded into each fuel channel depending on the model – a 790 MWe CANDU reactor contains 480 fuel channels composed of 5,760 fuel bundles containing over 5 million fuel pellets.

The on-load refueling is a fully-automated process: new fuel is inserted into a channel at one end and used fuel is collected at the other. This feature means that the PHWR is inherently flexible with its fuel requirements, and can run on different fuels requiring different residence times, eg natural uranium, slightly enriched uranium, plutonium-bearing fuels and thorium-based fuels.

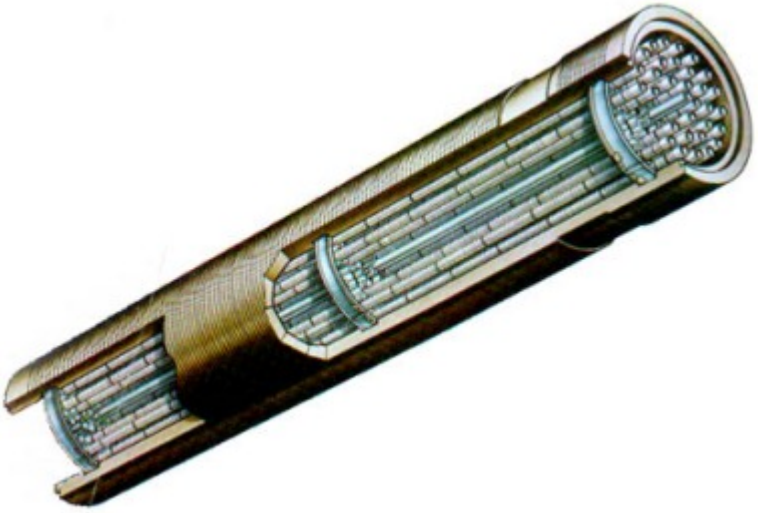
Figure 7: Indian PHWR fuel bundles

AGR fuel

The advanced gas-cooled reactor (AGR) is a second-generation UK-designed nuclear reactor only used in UK. AGRs account for about 2.7% of total global nuclear generating capacity. They employ a vertical fuel channel design, and use CO_2 gas – a very weak moderator – as the primary coolant.

AGR fuel assemblies consist of a circular array of 36 stainless steel clad fuel pins each containing 20 enriched UO_2 fuel pellets, and the assembly weighs about 43 kilograms. Enrichment levels vary up to about 3.5%. Stainless steel allows for higher operating temperatures but sacrifices some neutron economy. The assembly is covered with a graphite sheath which serves as a moderator. Eight assemblies are stacked end on end in a fuel channel, inserted down through the top of the reactor. During refueling this whole stack is replaced. Fuel life is about five years, and refueling can be carried out on-load through a refuelling machine.

Figure 8: cutaway of an AGR fuel assembly



RBMK fuel

The RBMK reactor is an early Soviet design, developed from plutonium production reactors. Eleven units are in operation (3% of world total), with control systems and oxide fuel greatly modified since 1990. It employs vertical pressure tubes (just under1700 of these, each about 7 metres long) running through a large graphite moderator. The fuel is cooled by light water water, which is allowed to boil in the primary circuit, much as in a BWR.

RBMK fuel rods are about 3.65 metres long, and a set of 18 forms a fuel bundle about 8 cm diameter. Two bundles are joined together and capped at either end by a top and bottom nozzle, to form a fuel assembly with an overall length of about 10 metres, weighing 185 kilograms. Since 1990 RBMK fuel has had a higher enrichment level, increasing from about 2% to average 2.8% (varying along the fuel element from 2.5% to 3.2%) and it now includes about 0.6% erbium (a burnable absorber). This has the effect of improving overall safety and increasing fuel burn-up. This new fuel can stay in the reactor for periods of up to six years before needing to be

removed. All RBMK reactors now use recycled uranium from VVER reactors.

As with other pressure tube designs such as the PHWR, the RBMK reactor is capable of on-load refueling.

Fast neutron reactor fuel

There is only one commercially operating fast reactor (FNRs) in service today – the BN-600 at Beloyarsk in Russia. There are two FNRs under construction – a 800 MWe unit in Russia and a 500 MWe unit in India (which expects to build five more). Two BN-800 units were planned in China.

Fast neutron reactors (FNR) are unmoderated and use fast neutrons to cause fission. Hence they mostly use plutonium as their basic fuel, or sometimes high-enriched uranium to start them off (they need about 20-30% fissile nuclei in the fuel). The plutonium is bred from U-238 during operation. If the FNR is configured to have a conversion ratio above 1 (ie more fissile nuclei are created than fissioned) as originally designed, it is called a Fast Breeder Reactor (FBR). FNRs use liquid metal coolants such as sodium and operate at higher temperatures. (See also Fast Neutron Reactor information paper)

Apart from the main FNR fuel, there are numerous heavy nuclides - notably U-238, but also Am, Np and Cm that are fissionable in the fast neutron spectrum – compared with the small number of fissile nuclides in a slow (thermal) neutron field (just U-235, Pu-239 and U-233). A FNR fuel can therefore include a mixture of transuranic elements. Also it can be in one of several chemical forms, including; standard oxide ceramic, mixed oxide ceramic (MOX), single or mixed nitride ceramics, carbides and metallic fuels. Further, FNR fuel can be fabricated in pellet form or using the 'vibro-pack' method in which graded powders are loaded and compressed directly into the cladding tube. Carbide fuels such as used in India have a higher thermal conductivity than oxide fuels and can attain breeding ratios larger than those of oxide fuels but less than metal fuels.

The core of an FNR is much smaller than a conventional reactor, and cores tend to be designed with distinct 'seed' and 'blanket' regions according to whether the reactor is to be operated as a 'burner' or a 'breeder'. In each case the fuel composition for the seed and blanket regions are different – the central seed region uses fuel with a high fissile content (and thus high power and

neutron emission level), and the blanket region has a low fissile content but a high level of neutron absorbing material which can be fertile (for a breeding design, eg U-238) or a waste absorber to be transmuted.

BN-600 fuel assemblies are 3.5 m long, 96 mm wide, weigh 103 kg and comprise top and bottom nozzles (to guide coolant flow) and a central fuel bundle. The central bundle is a hexagonal tube and for seed fuel houses 127 rods, each 2.4 m long and 7 mm diameter with ceramic pellets in three uranium enrichment levels; 17%, 21% and 26%. Blanket fuel bundles have 37 rods containing depleted uranium. BN-600 fuel rods use low-swelling stainless steel cladding.

FNRs use liquid metal coolants such as sodium or a lead-bismuth eutectic mixture and these allow for higher operating temperatures – about 550°C, and thus have higher energy conversion efficiency. They are capable of high fuel burn-up.

Nuclear fuel performance in reactors

Nuclear fuel operates in a harsh environment in which high temperature, chemical corrosion, radiation damage and physical stresses may attack the integrity of a fuel assembly. The life of a fuel assembly in the reactor core is therefore regulated to a burn-up level at which the risk of its failure is still low. Fuel 'failure' refers to a situation when the cladding has been breached and radioactive material leaks from the fuel ceramic (pellet) into the reactor coolant water. The radioactive materials with most tendency to leak through a cladding breach into the reactor coolant are fission-product gases and volatile elements, notably; krypton, xenon, iodine and cesium.

Fuel leaks do not present a significant risk to plant safety, though they have a big impact on reactor operations and (potentially) on plant economics. For this reason, primary coolant water is monitored continuously for these species so that any leak is quickly detected. The permissible level of released radioactivity is strictly regulated against specifications which take into account the continuing safe operation of the fuel. Depending on its severity a leak will require different levels of operator intervention:

1. Very minor leak: no change to operations – the faulty fuel assembly with leaking rod(s) is removed at next refuelling, inspected, and possibly re-loaded.

2. Small leak: allowable thermal transients for the reactor are restricted. This might prevent the reactors from being able to operate in a 'load-follow' mode and require careful monitoring of reactor physics. The faulty fuel assembly with leaking rod is generally removed and evaluated at the next scheduled refuelling.
3. Significant leak: the reactor is shut down and the faulty assembly located and removed.

A leaking fuel rod can sometimes be repaired but it is more usual that a replacement assembly is needed (this having a matching level of remaining enrichment). Replacement fuel is one cost component associated with failed fuel. There is also the cost penalty and/or replacement power from having to operate at reduced power or having an unscheduled shutdown. There may also be higher operation and maintenance costs associated with mitigating increased radiation levels in the plant.

Fuel management is a balance between the economic imperative to burn fuel for longer and the need to keep well within failure-risk limits. Improving fuel reliability extends these limits, and therefore is a critical factor in providing margin to improve fuel burn-up.

The nuclear industry has made significant performance improvements reducing fuel failure rates by about 60% in the 20 years to 2006 to an average of some 14 leaks per million rods loaded [IAEA 2010]. The reliability drive continues. Industry-wide programs led by the Electric Power Research Institute (EPRI) and the US Institute of Nuclear Power Operations (INPO) have produced guidelines to help eliminate fuel failures (there was an ambitious goal to achieve zero fuel failures by 2010). These programs led to the accident-tolerant fuel program (see below).

Fuel failures in US power reactors are rare. As of early 2014, 97% of US nuclear power plants were free of fuel failures, compared with 71% in 2007, according to EPRI. The annual US failure rate is about one in one million (*i.e.* five rods per year). Fuel engineering continues to improve, *e.g.* with more sophisticated debris filters in assembly structures. Utilities themselves impose more rigorous practices to exclude foreign material entering primary coolant water. Global Nuclear Fuel (GNF) in 2013 had two million fuel rods in operation and claimed to have no leakers among them. (In the early 1970s hydriding and pellet-clad interaction caused a lot of leaks. The 1980s saw an order of magnitude improvement.)

At the same time there has been a gradual global trend toward higher fuel burnup*. However, higher burnup generally requires higher enrichment levels, and there is a limit on this given the strict criticality safety limitation imposed on fuel fabrication facilities – the

maximum uranium enrichment level that can be handled is normally 5% U-235.

* Higher burnup does not necessarily mean better energy economics. Utilities must carefully balance the benefits of greater cycle length against higher front-end fuel costs (uranium, enrichment). Refuelling outage costs may also be higher, depending on length, frequency and the core re-load fraction.

An equally important trend in nuclear fuel engineering is to be able to increase the power rating for fuels, ie, how much energy can be extracted per length of fuel rod. Currently this is limited by the material properties of the zirconium cladding.

Fabrication supply and demand

The current annual demand for LWR fuel fabrication services is expressed as a requirement for about 7000 tonnes of enriched uranium being made into assemblies, and this is expected to increase to about 9500 t by 2020. Requirements for PHWRs account for an additional 3,000 t/year and the Gas-cooled Reactor market for around 400 t/year.

Requirements for fuel fabrication services will grow roughly in line with the growth in nuclear generating capacity. However, fabrication requirements are also affected by changes in utilities' reactor operating and fuel management strategies, which are partly driven by technical improvements in fuel fabrication itself. For example, LWR discharge burn-ups have increased steadily as improvements in fuel design have made this possible, and this has tended to reduce fabrication demand, as fuel remains in the reactor for a longer period (though there is a limit to how far burn-ups can be pushed without tackling the 5% enrichment limit in place for criticality safety margins at fuel fabrication plants). A parallel industry-wide focus on increasing fuel performance and reliability has also decreased the demand for fuel to replace defective assemblies.

Plans to build many new reactors affect the demand on fabrication capacity in two ways. The demand for reloads increases in line with the new installed reactor capacity, typically 16 to 20 tonnes per year per GW. Additionally the first cores create a temporary peak demand, since the amount required is about three to four times that of a reload batch in currently operated LWRs (and some of the enrichment is less). An average first core enrichment is about 2.8%.

Provision of fuel fabrication worldwide

Fuel fabrication services are not procured in the same way as, for example, uranium enrichment is bought. Nuclear fuel assemblies are highly engineered products, made to each customer's individual specifications. These are determined by the physical characteristics of the reactor, by the reactor operating and fuel cycle management strategy of the utility as well as national, or even regional, licensing requirements.

Most of the main fuel fabricators are also reactor vendors (or owned by them), and they usually supply the initial cores and early reloads for reactors built to their own designs. However the market for LWR fuel has become increasingly competitive and for most fuel types there are now several competing suppliers – most notably perhaps, Russian fabricator TVEL competes to manufacture Western PWR fuel, and Western fuel fabricators can manufacture VVER fuel. Early in 2016, 41% of Ukraine's VVER fuel came from Westinghouse in Sweden. In May 2016 Global Nuclear Fuel – Americas agreed with TVEL to produce its TVS-K fuel design in the USA for Westinghouse PWRs. TVEL also plans to market the fuel in Europe, and has been qualifying lead assemblies at Ringhals in Sweden.

Currently, fuel fabrication capacity for all types of LWR fuel throughout the world considerably exceeds the demand. It is evident that fuel fabrication will not become a bottleneck in the foreseeable supply chain for any nuclear renaissance. The overcapacity is increased by countries such as China, India and South Korea aiming to achieve self-sufficiency.

In May 2014 a European Commission staff report suggested that as a condition of investment, any non-EU reactor design built in the EU should have more than one source of fuel. The EC's May 2014 European Energy Security Strategy urged: "Ideally, diversification of fuel assembly manufacturing should also take place, but this would require some technological efforts because of the different reactor designs." In June 2015 the Euratom Research and Training Programme provided €2 million to Westinghouse and eight European partners "to establish the security of supply of nuclear fuel for Russian-designed reactors in the EU," especially VVER-440 types. Conceptual design was completed in May 2017, based on fuel provided by Westinghouse to Loviisa in 2001-07.

LWR fuel fabrication capacity worldwide is shown in Table 1. The back-conversion capacities are particularly unevenly distributed. For some fabricators it represents a bottleneck. Some fabricators do

not have conversion facilities at all and have to buy such services in the market, while others with excess capacity are even sellers of UO_2 powder.

Table 1: World LWR fuel fabrication capacity, tonnes/yr

	Fabricator	Location	Conversion	Pelletizing	Rod/assembly
Brazil	INB	Resende	160	120	400
China	CJNF Jianzhong	Yibin	800	800	800
	CBNF	Baotou	0	0	400
	CNNFC	Baotou	200	200	200
France	Framatome-FBFC	Romans	1800	1400	1400
	Orano	Malvési	Under const.		
Germany	Framatome-ANF	Lingen	800	650	650
India	DAE Nuclear Fuel Complex	Hyderabad	48	48	48
Japan	NFI (PWR)	Kumatori	0	383	284
	NFI (BWR)	Tokai-Mura	0	250	250
	Mitsubishi Nuclear Fuel	Tokai-Mura	450	440	440
	Global Nuclear Fuel – Japan	Kurihama	0	620	630
Kazakhstan	Ulba	Ust Kamenogorsk	0	108	200
Korea	KNFC	Daejeon	700	700	700
Russia	TVEL-MSZ*	Elektrostal	1500	1500	1560
	TVEL-NCCP	Novosibirsk	450	1200	1200
Spain	ENUSA	Juzbado	0	500	500
Sweden	Westinghouse AB	Västeras	787	600	600
UK	Westinghouse**	Springfields	950	600	860
USA	Framatome Inc	Richland	1200	1200	1200
	Global Nuclear Fuel – Americas	Wilmington	1200	1000	1000

	Fabricator	Location	Conversion	Pelletizing	Rod/assembly
	Westinghouse	Columbia	1600	1594	2154
Total			12,645	13,913	15,476

* Includes approx. 220 tHM for RBMK reactors
** Includes approx. 200 tHM for AGR reactors

Source: World Nuclear Association Nuclear Fuel Report 2021, Table 8.2
NB the above figures are about 40% above operational capacities, which meet demand.

Table 2: World PHWR fuel fabrication capacity, tonnes/yr

	Fabricator	Location	Rod/Assembly
Argentina	CONUAR	Cordoba & Eizeiza	160
Canada	Cameco	Port Hope	1500
	GNF-Canada	ToronPeterborough	1500
China	CNNFC	Baotou	246
India	DAE Nuclear Fuel Complex	Hyderabad	1000
Pakistan	PAEC	Chashma	20
Korea	KEPCO	Dejeon	800
Romania	SNN	Pitesti	250
Total			5476

Source: World Nuclear Association Nuclear Fuel Report 2021, table 8.3, from IAEA

The LWR fuel fabrication industry has rationalized in recent years, including:

- When Westinghouse Electric was bought by Toshiba, Kazatomprom acquired a 10% share of this (subsequently sold to Toshiba).

- Global Nuclear Fuels was formed as a joint venture between General Electric, Toshiba and Hitachi, though Toshiba sold its 14% stake to Hitachi in 2018, raising its share to 40%. There

are two 'branches' GNF-A (Americas) and GNF-J (Japan) with different ownership structures. It is best-known for BWR fuel.

- Toshiba purchased 52% of Nuclear Fuel Industries (NFI) in Japan, then agreed to buy the balance from Sumitomo (24%) and Furukawa (24%) to make it wholly-owned.
- Mitsubishi Heavy Industries and AREVA (30%) bought into Mitsubishi Nuclear Fuel and created a US fuel fabrication joint venture.
- Kazatomprom and AREVA agreed to build a 1200 t/yr fuel fabrication plant in Kazakhstan.*

* Kazatomprom has said that it aims to supply up to one third of the world fuel fabrication market by 2030, with China to be an early major customer.

Secondary supply from recycle

Currently about 100 t/yr year of reprocessed uranium (RepU) is produced at MSZ in Elektrostal, Russia (capacity 250 t/yr) for AREVA contracts. One production line in AREVA's plant in Romans, France is licensed to fabricate 150 t of RepU into fuel per year and PWR assemblies of this type have already been delivered to French, Belgian and UK reactors, and an amount of RepU powder has been sent from Russia to Japan. Limited RepU and enriched RepU (ERU) capacity exists elsewhere also.

At present, nearly all commercial MOX fuel is fabricated in AREVA's MELOX plant in Marcoule. With a capacity of 195 tonnes/yr and a good production rate this plant helps not only to save uranium and enrichment demand, but also frees up LWR fabrication capacity in the market.

The UK's Sellafield MOX plant had a designed capacity of 120 t/yr but was downgraded to 40 t/yr and never reached that level of reliable output before being closed down in 2011. Russia's MOX plant at Zheleznogorsk for fast reactors commenced operation in 2015. Japan's Rokkasho-Mura MOX plant is planned to be operational by 2022, and the US MOX plant in Savannah River was due to produce MOX fuel from weapons plutonium but this project has now been terminated.

The MOX fuel market has weakened somewhat recently with the cessation of its use in Belgium, Germany and Switzerland

(moratorium), and the continued loading of MOX fuel in Japan has diminished in the aftermath of the Fukushima accident.

Table 3: World MOX fuel fabrication capacity, tonnes/yr

	Fabricator	Location	Pelletising	Rod/assembly
France	Orano	Marcoule	195	195
India	DAE Nuclear Fuel Complex	Tarapur	50	50
Japan	JAEA	Tokai-Mura	5	5
	JNFL	Rokkasho-Mura*	130	130
Russia	MCC	Zheleznogorsk	60	60
Total			440	440

* Operational by 2022

Source: World Nuclear Association Nuclear Fuel Report 2019, Table 8.7, updated

MOX fuel

Mixed uranium oxide + plutonium oxide (MOX) fuel has been used in about 30 light-water power reactors in Europe and about ten in Japan. It consists of depleted uranium (about 0.2% U-235), large amounts of which are left over from the enrichment of uranium, and plutonium oxide that derives from the chemical processing of used nuclear fuel (at a reprocessing plant). This plutonium is reactor-grade, comprising about one third non-fissile isotopes.

In a MOX fuel fabrication plant the two components are vigorously blended in a high-energy mill which intimately mixes them such that the powder becomes mainly a single 'solid solution' $(U,Pu)O_2$. MOX fuel with about 7% of rector-grade plutonium is equivalent to a typical enriched uranium fuel. The pressing and sintering process is much the same as for UO_2 fuel pellets, though some plastic shielding is needed to protect workers from spontaneous neutron emissions from the Pu-240 component.

Vibropacked MOX (VMOX) fuel is a Russian variant for MOX fuel production in which blended $(U,Pu)O_2$ and UO_2 powders are directly loaded and packed into cladding tubes where they sinter in-situ under their own operating temperature. This eliminates the need to manufacture pellets to high geometric tolerances, which involves grinding and scrap which are more complex to deal with for Pu-

bearing fuels. Russian sources say vibropacked fuel is more readily recycled.

REMIX fuel

REMIX (Regenerated Mixture) fuel is produced directly from a non-separated mix of recycled uranium and plutonium from reprocessing used fuel, with a LEU (up to 17% U-235) make-up comprising about 20% of the mix. This gives fuel with about 1% Pu-239 and 4% U-235 which can sustain burn-up of 50 GWd/t over four years. It is distinct from MOX fuel in having low and incidental levels of plutonium – none is added. The spent REMIX fuel after four years is about 2% Pu-239* and 1% U-235, and following cooling and reprocessing the non-separated uranium and plutonium is recycled again after LEU addition, which compensates for the even isotopes of both elements.

REMIX fuel can be repeatedly recycled with 100% core load in current VVER-1000 reactors and, correspondingly reprocessed many times – up to five times, so that with three fuel loads in circulation a reactor could run for 60 years using the same fuel, with LEU recharge. REMIX can serve as a replacement for existing reactor fuel and Rosatom is now proceeding to pilot operation of several full-REMIX fuel assemblies. In August 2020 it announced that REMIX fuel for VVER-1000 reactors would be produced on a new production line at the Siberian Chemical Plant (SCC) at Seversk from 2023.

* a 68% increase, compared with 104% in MOX fuel cycle, according to Tenex.

TRISO high temperature reactor fuel

High temperature reactors (HTR) operate at 750 to 950°C, and are normally helium-cooled. Fuel for these is in the form of TRISO (tristructural-isotropic) particles less than a millimetre in diameter. Each has a kernel (ca. 0.5 mm) of uranium oxycarbide (or uranium dioxide), with the uranium enriched up to 20% U-235, though normally less. This is surrounded by layers of carbon and silicon carbide, giving a containment for fission products which is stable up to very high temperatures. Trials at two US laboratories have confirmed that most fission products remain securely in TRISO particles up to about 1800°C.

There are two ways in which these particles can be arranged in a HTR: in blocks – hexagonal 'prisms' of graphite; or in billiard ball-sized pebbles of graphite encased in silicon carbide, each with about 15,000 fuel particles and 9g uranium. Either way, the moderator is graphite. HTRs can potentially use thorium-based fuels, such as high-enriched or low-enriched uranium with Th, U-233 with Th, and Pu with Th. Most of the experience with thorium fuels has been in HTRs.

The main HTR fuel fabrication plant is at Baotou in China, the Northern Branch of China Nuclear Fuel Element Co Ltd. From 2015 this makes 300,000 fuel pebbles per year for the HTR-PM under construction at Shidaowan. Previous production has been on a small scale in Germany.

In the USA, BWX Technologies at Lynchburg in Virginia is making high-assay low-enriched (HALEU) TRISO fuel on an engineering scale, funded by the US Department of Energy (DOE), and in October 2019 the company announced an expansion to commercial scale within three years. In March 2020 the DOE awarded a contract to BWXT to fabricate HALEU TRISO fuel to support development of the DOE's Transformational Challenge Reactor (TCR) project. The HTR, reported to be 3 MWt, is to be built at Oak Ridge National Laboratory (ORNL) in Tennessee to demonstrate reduced deployment costs "using a rapid advanced manufacturing approach." The DOE intends the reactor to be designed and built using 3D printing, and achieve criticality by 2023. "An agile approach to design, manufacturing, and testing is employed to meet this schedule and to deliver a new paradigm to designing and deploying nuclear systems." The reactor core consists of uranium nitride coated fuel particles within an advanced manufactured silicon carbide structure. The fuel blocks are interspersed with yttrium hydride moderator elements.

X-energy has a TRISO pilot fuel fabrication facility at ORNL. In November 2019 X-energy and GNF agreed to set up commercial HALEU TRISO production at GNF's Wilmington plant in North Carolina. This is expected to produce TRISO fuel of "significantly higher quality and at costs that are substantially lower than other potential manufacturers." It would potentially supply the US Department of Defense for micro-reactors and NASA for nuclear thermal propulsion. X-energy is building on the TRISO fuel technology developed under the DOE's Advanced Gas Reactor Fuel Qualification Program through two cooperative agreements with the DOE. In October 2020 the DOE awarded X-energy a grant under the Advanced Reactor Demonstration Program which involved building a TRISO fuel plant.

X-energy also has agreements with Centrus Energy in the USA to develop TRISO fabrication technology for uranium carbide fuel, and with NFI at Tokai in Japan, which has 400 kgU/yr HTR fuel capacity, for Japan's 30 MWt HTTR. NFI is to supply equipment for X-Energy's TRISO-X plant at ORNL, which may involve relocating the whole plant.

X-energy is applying for a loan guarantee from the government for commercialization of a TRISO-based fuel supply chain and is expected to submit a licence application for a commercial plant by mid-2021, though this may now be GNF's prerogative.

Other high-assay low-enriched fuel

In connection with a number of small modular reactor (SMR) designs, attention is turning to the need for high-assay low-enriched uranium (HALEU), with enrichment levels between 5% and 20% U-235. In the USA, the Department of Energy (DOE) is proposing to convert metallic HALEU into fuel for research and development purposes at Idaho National Laboratory's Materials and Fuels Complex and/or the Idaho Nuclear Technology and Engineering Center, to support the development of new reactor technologies with higher efficiencies and longer core lifetimes. HALEU may be metallic or oxide.

HALEU can be produced with existing centrifuge technology, but a number of arrangements would need to be made for this, as well as for deconversion and fuel fabrication. New transport containers would also be required as those for today's enriched UF_6 could not be used due to criticality considerations.

Advanced nuclear fuel technology

Fuel development activities in the nuclear industry have largely focused on improving the reliability of standard zirconium-clad uranium oxide fuels. Increasingly, however, R&D effort is being applied to evolutionary fuel forms that can offer significant improvements in terms of safety, waste management and operating economics, as well as allowing the deployment of new types of reactor.

Accident tolerant fuel

Accident tolerant fuel (ATF) is a term used to describe new technologies that enhance the safety and performance of nuclear fuel. ATF may incorporate the use of new materials and designs for cladding and fuel pellets.

Framatome, GE/GNF and Westinghouse are all developing ATF concepts with the help of funding from the US Department of Energy (DOE). Since 2012, the DOE has supported the development of ATF concepts through its Enhanced Accident Tolerant Fuel (EATF) programme. Its objective is to develop new cladding and fuel materials that can better tolerate the loss of active cooling in the core, while maintaining or improving fuel performance and economics during normal operations. A priority of the EATF programme is to minimise the generation of hydrogen.

The joint DOE programme uses the Halden research reactor in Norway to test ATF fuel rods, as well as the Advanced Test Reactor (ATR) and the restarted Transient Reactor Test Facility (TREAT) at the DOE's Idaho National Laboratory (INL). In February 2017 the DOE awarded $10 million to Framatome (then Areva) over two years for phase 2 of the programme, and similar funding is being provided for GE Hitachi and Westinghouse.

Framatome in phase 2 of its PROtect enhanced ATF programme from 2017 has been developing a nuclear fuel concept, using a chromium-coated zirconium alloy cladding (M5) combined with chromia-doped fuel pellets. The fuel is expected to retain fission gases better and improve pellet-cladding interaction, and the cladding will better resist high-temperature oxidation. The GAIA spacer grid holding the fuel rods also has high mechanical fretting resistance. In June 2018 the DOE announced testing of Framatome's ATF in the Advanced Test Reactor at Idaho National Laboratory. The first full test assemblies of this GAIA fuel with M5 cladding and chromia-enhanced pellets were loaded into Southern Nuclear's Vogtle 2 in March 2019. Exelon plans to load two full GAIA fuel assemblies into Calvert Cliffs 2 in March 2021. Entergy is also due to use them in Arkansas 1. These will also have chromium-coated zircalloy cladding and chromia-doped fuel pellets. Framatome's main BWR advanced fuel is Atrium though in 2021 it supplied both PROtect and Atrium fuel to Monticello BWR, the eighth US reactor to change to Atrium. Framatome is also continuing work on a silicon carbide cladding, and plans to use that cladding on chromia-doped pellets in lead test assemblies in about 2022.

GE Hitachi with GNF is developing two types of ATF: a ferritic/martensitic steel alloy cladding (e.g. Fe-Cr-Al) known as

IronClad, and a coated zirconium cladding known as ARMOR. Both are for conventional UO_2 fuel and are designed to provide oxidation resistance and superior material behaviour over a range of conditions in BWRs. The IronClad Fe-Cr-Al cladding has better mechanical strength at high temperatures, retains fission gases better than zirconium alloy and has less potential for hydrogen generation in an accident. The ARMOR-coated zirconium cladding provides enhanced protection of fuel rods against debris fretting.

GNF's advanced fuel rods were the first developed through the ATF programme to be loaded into a commercial reactor during Southern Nuclear's Hatch 1 spring refuelling outage in March 2018. These were unfuelled IronClad lead test rods and fuelled lead test rods with ARMOR-coated zirconium cladding. After a full 24-month fuel cycle these were removed and sent to DOE's Oak Ridge National Laboratory for testing. Exelon's Clinton plant loaded lead test assemblies in January 2020, both with ARMOR-coated zirconium cladding and three varieties of GNF's IronClad, which are the first fuelled ferritic steel-based cladding assemblies to be installed in a commercial reactor.

Westinghouse in June 2017 launched its EnCore ATF. It is manufacturing the first EnCore test rods, with lead test assembly insertion of these in September 2019, at Exelon's Byron plant. The initial EnCore fuel comprises high-density uranium silicide fuel pellets inside zirconium cladding with a thin coating of chromium making it more robust chemically. (Uranium silicide – U_3Si_2 – fuels for research reactors are being developed at INL also.) In the second phase, the uranium silicide fuel pellets would be in silicon carbide ceramic matrix composite cladding with a melting point of 2800°C, and these test assemblies could be loaded into a reactor by 2022. The EnCore ATF fuel development has been supported by awards from the DOE to Westinghouse and a group of partners including General Atomics, several DOE national laboratories, Southern Nuclear Operating Company and Exelon.

Westinghouse said that cost savings will arise because the uranium silicide offers up to 20% higher density of uranium and much higher thermal conductivity which does not degrade with irradiation like UO_2, so fewer fuel assemblies need to be replaced during each refuelling outage. In the second phase of EnCore, the higher temperature tolerance of silicon carbide cladding has potential for revised regulatory requirements, and Westinghouse sees this as a "game changer".

After trials of lead test assemblies, Westinghouse intends to make full reload quantities available from 2027. ATFs present a number

of manufacturing challenges and, given the neutronic penalties entailed, enrichment to over 5% could be needed, despite the higher density of uranium in the fuel.

Rosatom's fuel company TVEL plans to offer ATF to its customers by the early 2020s. TVEL is developing ATF for use in Rosatom's VVER reactors and in Western PWRs. Testing of prototype assemblies – both VVER and one for Western PWR – commenced early in 2019 at the MIR research reactor at the State Research Institute of Atomic Reactors in Dimitrovgrad. Each assembly consists of 24 fuel rods with different combinations of cladding materials and fuel composition. After testing, the VVER ones passed inspection in December 2019. In September 2021 pilot operation of three ATF fuel assemblies commenced in Rostov 2 VVER-1000 reactor. Each of the three TVS-2M assemblies contains twelve ATF rods with cladding composed of either zirconium alloy with a heat-resistant chromium coating, or chrome-nickel alloy. Also in research reactors Rosatom will continue to irradiate fuel rods with various combinations of cladding and fuel pellet composition which may include uranium-molybdenum alloy or uranium disilicide. The Bochvar Institute is developing fabrication technology for uranium disilicide pellets. As well as high uranium density, it points to high thermal conductivity and low heat capacity of silicide fuel.

The OECD Nuclear Energy Agency's Expert Group on ATFs for Light Water Reactors reviews cladding and core materials focusing on their fundamental properties and behaviour under normal operation and accident conditions, as described above for the US-led programme. Both advanced core materials and components, in particular innovative cladding materials (coated and improved Zr-based alloys, SiC and SiC/SiC composites, advanced steels and refractory metals) and non-fuel components (advanced control rods, BWR channel box) are considered. A sub-group focuses on fuel design to address three categories of innovative fuels: improved UO_2, high-density fuels and coated-particle fuels such as the HTR fuel described above.

Metal fuel

Independently of the US DOE and the international ATF program, **Lightbridge** is developing an advanced metal fuel concept that may have accident tolerant characteristics.

Metal fuels were used in some earlier reactors such as the UK Magnox design, and also in two US fast reactors, with 5-10%

zirconium alloyed. But the higher melting point of uranium oxide has made it the preferred fuel in all reactors for half a century.* At least in the USA, metal fuels have not been made since the 1980s.

* UO_2 has a very high melting point – 2865°C (compared with pure uranium metal – 1132°C).

However, metal has much better thermal conductivity than ceramic oxide, and recent research has turned back to metal fuel forms. Babcock & Wilcox Nuclear Energy was working with Lightbridge to set up a pilot plant for metal fuel which is 50:50 (by mass) Zr-U alloy, with uranium enriched to almost 20% and having a multi-lobed and helically twisted rod geometry. The increased enrichment compensates for reductions in the initial fissile loading and in the derivative plutonium. Melting point of the alloy is about 1600°C, and average operating temperature in the fuel is up to 370°C (rather than about 1250°C in normal oxide fuel), the thermal conductivity being five times better than oxide fuel. BWXT in the USA has now completed its assessment of feasibility and prepared a fabrication plan for manufacturing fuel samples.

Each Lightbridge fuel rod consists of a central displacer of zirconium surrounded by a four-lobed fuel core with the cladding metallurgically bonded to it. For the hexagonal fuel assemblies for VVERs, the fuel rod is three-lobed. The shape of the rod provides increased surface area for heat transfer and the area between the lobes accommodated swelling during irradiation. The rod has greater structural integrity than current tubes with ceramic pellets inside. The twist of about 180° over about a metre means that the rods are self-spacing while giving good flow characteristics. The fuel operates at a higher power density than oxide fuels and the target burn-up is 21 atomic percent, about three times that of oxide fuels. It is suitable for all LWRs, and is expected to give a power uprate of about 17% in existing PWRs, and up to 30% in new ones designed for the higher power density, with longer fuel cycle. In adition to US and Russian patents, in June 2015 the design was patented in South Korea, where Lightbridge sees a “significant potential market”, and in July 2017 this patent protection was extended to cover both the metallic four-lobe design and its manufacture from powder. By November 2017 it had been patented in Japan, China (four patents), South Korea, and Canada.

Lightbridge has also agreed with Canadian Nuclear Laboratories for fabrication of such metal fuel at Chalk River in Canada and testing of it there in the NRU reactor. The agreement was expected to see fabrication and characterization of prototype fuel rods using depleted uranium in early 2016, with irradiation fuel samples using

enriched uranium made later the same year. Subject to final approval from the Norwegian Radiation Protection Authority, Lightbridge will test the fuel under prototypical PWR conditions in a pressurized water loop of Norway's 25 MWt Halden research reactor (a boiling heavy water reactor). The initial phase of irradiation testing was to begin in 2017 using short samples to evaluate conductivity, and continue for about three years using 70 cm fuel rods to evaluate cladding and swelling. Tests aim to reach the burnup necessary for insertion of lead test assemblies (LTAs) in a commercial power reactor. The final phase of irradiation testing necessary for batch reloads and full cores operating with a 10% power uprate and a 24-month cycle is expected to take an additional two years and be completed while LTAs have begun operating in the core of a commercial power reactor about 2020.

In April 2015 a group of electric utilities representing half of the US nuclear generating capacity wrote to the Nuclear Regulatory Commission formally expressing interest in Lightbridge's metal fuel, saying that they believed the fuel provided opportunities to improve safety and fuel cycle economics significantly. The utilities' Nuclear Utility Fuel Advisory Board (NUFAB) expects to test the fuel in an operating PWR about 2020.

In March 2016 Lightbridge entered into an exclusive joint development agreement with Areva NP to set up a 50:50 joint venture that would develop, fabricate and commercialize fuel assemblies based on the metallic fuel technology. In November it announced an agreement on key terms for the US-based joint venture, creating "a viable and well-defined commercialization path" covering fuel assemblies for most types of light water reactor, including pressurized water reactors (excluding VVERs), boiling water reactors, small modular reactors and research reactors. In September 2017 a binding agreement was signed with Areva Inc (for New NP) to set up the joint venture in North America. The joint venture between Lightbridge and Framatome* was officially launched in January 2018 and assigned the name Enfission. Commercial sales of the fuel were expected by 2026. However, in 2019 Lightbridge sought to terminate the joint venture and early in 2020 it was seeking new partners. In March 2021 the two companies agreed to dissolve the joint venture, with the IP rights for the fuel reverting to the source companies.

* New NP was renamed Framatome in January 2018.

Lightbridge is working with four US nuclear utilities, and late in 2016 a letter of intent was signed with one of them for a lead test

fuel assembly demonstration in a US commercial reactor, possibly by 2021.

In the USA the Idaho National Laboratory (INL) has been testing metal fuel fabrication by extrusion for TerraPower's so-called travelling wave reactor (TWR). The fuel in this is designed to be rearranged but not replenished for 40 years.

Thorium-uranium fuel under development

Since the early 1990s Russia has had a programme to develop a thorium-uranium fuel, based at Moscow's Kurchatov Institute and involving the US company Lightbridge Corporation and US government funding to design fuel for Russian VVER-1000 reactors. This is the Radkowsky Thorium Reactor concept. No recent progress with it is known.

Whereas normal fuel uses enriched uranium oxide throughout the fuel assembly, the new design has a demountable centre portion and blanket arrangement, with uranium-zirconium metal fuel rods in the centre and uranium-thorium oxide pellets in conventional fuel rods around it. The Th-232 in the blanket captures neutrons to become U-233, which is fissile and is fissioned in situ. Blanket material remains in the reactor for nine years but the centre portion is burned for only three cycles (as in a normal VVER, 3 or 4.5 years depending on the refuelling interval). No reprocessing of the blanket is envisaged, due both to the difficulty of doing so with thorium fuels and the presence of significant U-232 in the blanket. The two-part fuel assembly has the same geometry as a normal VVER one.

A variant of this design uses plutonium-zirconium metal fuel rods in the central seed assembly, and was earlier known as a plutonium incinerator.

Another thorium-uranium fuel is being developed in the USA by Clean Core Thorium Energy, using a mixture of HALEU and thorium for PHWR reactors such as CANDU. The company hopes to commercialize the ANEEL Fuel ('Advanced Nuclear Energy for Enriched Life') technology by 2025.

Other fuel developments

Other fuel technologies that seem particularly promising, and which could be commercially deployed in the foreseeable future include:

- Ceramic or coated zirconium claddings that prevent the adverse interaction between steam and zirconium at very high temperature

- High thermal conductivity oxide fuel, such as can be achieved by including additives like beryllium oxide (BeO). Higher conductivity provides higher safety margins and can allow higher operating powers.

- Thoria-based fuels, including mixed thorium-plutonium (Th-MOX) fuel which can achieve a high utilization factor for recycled plutonium.

- Other all-metal fuels and annular LWR fuel, allowing more cooling and therefore safe, high power densities for the fuel and improved economics.

- Pelletized coated-particle fuels, aimed at achieving high safety levels for the fuel that can be left in a light water reactor for very long periods, thereby achieving high burn-up of recycled plutonium and/or actinide waste components.

Work is underway on each of these new fuel technologies.

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